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ECONOMIC INTEGRATION AMONG DEVELOPED, DEVELOPING AND CENTRALLY PLANNED ECONOMIES: A COMPARATIVE ANALYSIS

Josef C. Brada and José A. Méndez*

Abstract—We examine six integration schemes and decompose their ability to increase inter-member trade into environmental, policy and system effects. Environmental factors caused the greatest variation in trade creation, with inter-member distance the most important environmental variable. The CACM and EFTA have followed more effective integration policies than the EEC, LAFTA and the Andean Pact. Although integration can thus benefit developed and developing countries alike, for some, such as those in Latin America, inter-member distances severely limit its effectiveness. While the combination of policy and system has kept the CMEA from achieving its full potential for increasing inter-member trade, its effectiveness does not differ from that of unions among market economies.

I. Introduction

THE ability of regional economic integration schemes to increase inter-member trade has been amply demonstrated for developed countries. The work of Aitken (1973) and Balassa (1967) showed that the formation of the EEC and EFTA significantly increased inter-member trade. Similarly, studies by Hewett (1976) and Pelzman (1977) showed that the Council for Mutual Economic Assistance (CMEA) increased inter-member trade among its East European members. Surprisingly, despite the existence of a number of regional integration schemes among developing countries, no similarly rigorous assessment of the impact of integration among developing countries has been attempted. This despite the interest of third-world leaders in increasing regional economic cooperation to counter developed-country protectionism and to promote export-oriented development strategies. Moreover, despite the evident success of the EEC and EFTA in increasing inter-member trade, little is known about the relative importance of the numerous factors contributing to this success. For example, the extent to which the success of these two schemes can be attributed to the type and extent of inter-member trade liberali-

zation undertaken and the extent to which it is the fortuitous result of the proximity or high level of development of the integrating countries are relatively unexplored issues despite their obvious relevance to the success of further efforts at regional economic integration.

In order to fill this lacuna, we provide estimates of the trade augmenting effects of three developing country integration schemes and compare them to those of three developed country schemes. The developing country schemes examined are the Central American Common Market (CACM), the Latin American Free Trade Area (LAFTA) and the Andean Pact. The developed country schemes to which we compare them are the EEC, EFTA and the CMEA.

II. The Model

With the exception of Balassa's (1967) study of trade creation and diversion in the EEC, the common approach to quantifying the effects of integration on trade flows has been to utilize the so-called gravity equation. The gravity equation has proven popular for several reasons. First it provides an empirically tractable general equilibrium framework for modelling bilateral trade flows. Second it has a sound theoretical basis (Bergstrand (1985)) and finally it has proved useful in a variety of applications (Geraci and Prewo (1977), Linnemann (1966), Poyhonen (1963) and Tinbergen (1962)). In its basic form the model is written as

$$\log X_{ij} = A + \alpha_1 \log Y_i + \alpha_2 \log Y_j + \alpha_3 \log N_i + \alpha_4 \log N_j + \alpha_5 \log D_{ij} + \log e_{ij} \quad (1)$$

where

X_{ij} = value of exports from country i to country j

A = constant

Y_i, Y_j = income in the exporting and importing countries

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N_i, N_j = population in the exporting and importing countries

D_{ij} = distance between countries i and j

e_{ij} = lognormal error term.

The income and population variables represent the trading countries' endowments and tastes. Since greater productive capacity and incomes promote trade, α_1 and α_2 are expected to be positive. Large countries have more diversified production and thus satisfy a greater proportion of domestic demand while small countries tend to be more specialized and thus more dependent on trade, suggesting that α_3 should be negative (Leamer and Stern (1970), pp 152–153). The population of the importing country should have a positive effect on the volume of trade, since a larger population permits a greater division of labor and diversity of production, enabling imports to compete with domestic goods at more stages of the production process. Moreover, a large market better compensates exporters for the cost of acquiring information and establishing a sales and distribution network. Thus α_4 should be positive.

The use of these variables to model the relationship between endowments, tastes and trade with a sample such as ours that includes both developed and developing countries raises some conceptual issues. The first is that we must assume that all countries in the sample are developing along similar lines and that the process of development will not alter their trade behavior in a way other than predicted by equation (1).¹ Second, aggregate measures of income differ between developed and developing countries both in the proportion of economic activity they are able to cover and in the mix of goods they include. Both biases operate in the same direction; developing country incomes are likely to be understated and each dollar of income includes fewer tradables than does a dollar of developed country income. To the extent that such a bias exists, it is offset by the log-linear specification of equation (1), which weighs each dollar of developing country income more than a dollar of a high income country.² Moreover, in our

measurement of environmental effects on integration we attempt to adjust for these differences between developed and developing countries.

The distance variable represents resistance to trade. This resistance has an economic element, consisting of transportation and information costs; a structural element reflecting differences in consumption patterns and resource endowments as, for example, between temperate and tropical countries; and a policy element including the effects of economic integration. Because the structural factors are ambiguous in their effect, with differences in endowments promoting trade but differences in consumption hindering it, we focus our attention on the other two factors by improving on the way that the effects of integration on resistance to trade are specified.

Researchers have used the gravity equation to measure the trade-augmenting effects of integration in two ways. The more common, used by Tinbergen (1962), Aitken (1973) and Hewett (1976), is to employ a set of dummy variables to measure the effect of integration on inter-member trade. Whenever two members of a preferential trading group trade with each other, the dummy is equal to 2; when trade is with or among non-members the dummy equals 1. The larger the value of the coefficient of the dummy variable, the greater the volume of inter-member trade relative to "normal" or non-preferential trade and the more effective the integration scheme. An alternative approach, employed by Pelzman (1977), is to choose a pre-integration period on the basis of which equation (1) is estimated. The parameter estimates are then employed to project expected inter-member trade during the post-integration period. The excess of actual inter-member trade over the expected volume of trade is attributed to the effect of integration. While either approach is acceptable when one or a few integration schemes, made up of countries at similar levels of development, of similar size and with the same economic system, are analyzed, both methods break down when we turn to the more heterogeneous sample of countries required to compare the effects of integration among developed and developing countries and among market and planned economies.

¹ Chenery (1960) presents evidence, based on a sample of 62 countries, suggesting that this assumption is acceptable.

² Because we uncover policy influences on the resistance variable that are correlated to the developed–developing country division of our sample any test of the appropriateness of pooling involving the entire sample would be biased against pooling. However, a test based on trade with the non-integrat-

ing countries in our sample suggests that pooling developed and developing countries is acceptable in the gravity equation framework.

The effect of integration on inter-member trade is influenced by three sets of factors. The first of these is the environment, which we take to mean the physical and economic characteristics of the integrating countries and their economic relations with the rest of the world. For example, countries close to each other should experience, *ceteris paribus*, a greater post-integration stimulus to mutual trade than would two integrating countries that are far from each other. A second influence on the effectiveness of integration is the economic system of the integrating countries. The literature on the trade behavior of the planned economies suggests that such economies will trade less, *ceteris paribus*, than comparable market economies. Finally, there is the element of policy; some integration schemes lower barriers against inter-member trade to a greater degree than do others and thus are more effective in increasing inter-member exchanges.³ When we deal with a homogeneous group of countries, we can assume that the integration dummies or the difference between actual and predicted trade flows do not reflect systemic differences; nor do we expect that environmental factors change sharply over time or differ appreciably between integration schemes. Thus the coefficients of the integration dummies or the differences between expected and actual post-integration trade can safely be attributed to the policy variable, economic integration. However, with a more heterogeneous sample, estimates of the effect of integration will become tainted at best and swamped at worst by the differences in system and environment that exist among the various integration schemes.

In order to overcome these difficulties, we modify the gravity equation to take into account environmental effects on the effectiveness of integration. The two environmental variables we model are distance among integrating countries and their level of development. The hypothesis that distance among integrating countries will influence the amount by which inter-member trade increases is based on many of the same arguments employed to include distance as a trade resistance variable. For example, if countries A, B and C agree to reduce trade barriers among themselves, and A and B are close to each other but both are distant

from C then the degree of integration and consequently the increase in trade should be greater between A and B than between A and C or B and C. In part this will occur because for some bulky or highly perishable products trade between A and B might become feasible while remaining uneconomic between A and C. The larger distance between A and C will also place businesses in C at a disadvantage vis-à-vis those in B in assessing and reacting to market opportunities in country A, since they will have less precise information about A and less direct acquaintance with its culture and economy. Thus they will be less successful in penetrating A's markets. Finally, countries close to each other are likely to have greater cultural similarities and also relatively similar climates, leading to similar patterns of consumption and production that, in turn, lead to greater opportunities for exchanges of products (Linder, 1961). Consequently, an integration scheme with relatively small distances among members should stimulate inter-member trade more, *ceteris paribus*, than one consisting of countries located far from each other.

The level of development should also have a positive impact on the effects of integration largely because less developed countries have a structural bias against trade, and thus benefit less from integration. Their production is concentrated in subsistence agriculture and in services, neither of which enter into international trade. The bulk of their trade is thus with countries of differing levels of development and consists of exchanges of agricultural products and raw materials for manufactures. Developed countries' production, concentrated in manufactures, permits both complementary trade (manufactures for raw materials) as well as a large measure of intra-industry exchanges generally not available to developing countries.

To measure these environmental influences on trade flows we rewrite equation (1) as

$$\begin{aligned} \log X_{ij} = & A + \alpha_1 \log Y_i + \alpha_2 \log Y_j + \alpha_3 \log N_i \\ & + \alpha_4 \log N_j + \alpha_5 \log D_{ij} + \beta \log Q_{ij} \\ & + \gamma_1 P_{ij} \log (Y_i/N_i)(Y_j/N_j) \\ & + \gamma_2 P_{ij} \log D_{ij} + \log e_{ij} \end{aligned} \quad (2)$$

where $Q_{ij} = 2$ and $P_{ij} = 1$ if countries i and j belong to the same preference area and 1 and 0, respectively, when countries i and j belong to different or no preference areas. The coefficient γ_1 measures the effect of per capita income on the

³ For a detailed presentation of the environment-system-policy framework, see Koopmans and Montias (1971).

effectiveness of integration. If the coefficient is positive, then the effect of integration on inter-member trade increases with the level of development of the integrating countries, reflecting the higher proportion of tradables in their output. The coefficient γ_2 measures the effect of distance on the trade augmenting power of a customs union. The greater the distance among members, the smaller, *ceteris paribus*, is the augmentation of their trade with each other.

III. Empirical Results

Data were collected for the trade of the member countries of EEC, EFTA, CACM, and LAFTA and the Andean Pact with each other and with eighteen developed and developing countries belonging to no integration scheme.⁴ Trade flows of the CMEA countries were not employed in estimating the parameters of equation (2) since, due to systemic differences between them and the other countries in the sample, the trade of the CMEA was not expected to follow the regime implied by parameters estimated on the basis of the trade of market economies.

Because the observations could not be pooled over time, it was necessary to estimate parameters for equation (2) for each year. Due to space constraints we present results for the last year for which complete data were available, 1976, and two preceding years, 1970 and 1973. This period corresponds to the years in which all of the integration schemes were in existence. The parameter esti-

TABLE 1.—PARAMETER ESTIMATES FOR EQUATION (2)
(*t*-ratios in parentheses)

Parameter	Year		
	1970	1973	1976
A	0.0974	2.711	1.606
α_1	1.092 (15.80)	0.972 (15.92)	1.034 (17.98)
α_2	0.157 (3.94)	0.136 (3.95)	0.146 (4.15)
α_3	-0.291 (-3.05)	-0.089 (-1.02)	-0.185 (-2.42)
α_4	0.574 (9.57)	0.477 (8.41)	0.442 (8.57)
α_5	-0.543 (-8.09)	-0.581 (-8.64)	-0.472 (-7.39)
β	3.772 (1.85)	4.679 (1.87)	4.831 (1.97)
γ_1	0.194 (2.13)	0.104 (1.85)	0.058 (0.83)
γ_2	-0.619 (-4.40)	-0.630 (-4.91)	-0.525 (-4.25)
R^2	0.651	0.693	0.706
N	864	774	789

mates are presented in table 1. The coefficients for income and population have the expected signs, and these as well as α_5 , the coefficient for distance, are similar to those reported by Aitken and Hewett, whose samples were restricted to industrialized countries. With the exception of the constant term and α_3 , the coefficients are relatively stable over time.

Of the coefficients measuring the effects of integration, β is positive, indicating that integration does reduce the resistance to trade among member countries. The coefficient for per capita incomes, γ_1 , is positive indicating that, *ceteris paribus*, integration among high per capita income countries causes a greater increase in trade than does integration among low-income countries. The value of γ_1 falls over time and in 1976 is not significantly different from zero. This we attribute to the effects of the worldwide increase in the prices of fuels and raw materials, since this increase then caused complementary trade in such goods among countries of different income levels to be weighted more heavily in total trade than competitive trade flows among developed countries.⁵ Finally, γ_2 , the coefficient of the distance dummy, is negative. This

⁴ The countries in the sample, with dates they joined (and left) a particular integration scheme are:

CACM: Guatemala (1961), El Salvador (1961), Honduras (1961/1970), Nicaragua (1962), Costa Rica (1963).

LAFTA: Mexico (1960), Argentina (1960), Brazil (1960), Paraguay (1960), Uruguay (1960), Bolivia* (1967), Chile* (1960), Colombia* (1961), Ecuador* (1961), Peru* (1960), Venezuela* (1965).

Andean Pact: The members of LAFTA marked by an asterisk joined the Andean Pact in 1970, save Venezuela which joined in 1974.

CMEA: Bulgaria, Czechoslovakia, East Germany, Hungary, Poland, Romania, USSR. The CMEA was officially formed in 1949.

EEC: Belgium-Luxembourg (1959), France (1959), Italy (1959), Netherlands (1959), Federal Republic of Germany (1959), Denmark (1973), United Kingdom (1973).

EFTA: Austria (1960), Denmark (1960/1973), Norway (1960), Portugal (1960), Sweden (1960), Switzerland (1960), United Kingdom (1960/1973).

Non-members: Algeria, Egypt, Zaire, Nigeria, South Africa, Iraq, Iran, Saudi Arabia, Hong Kong, Singapore, Japan, Canada, Panama, United States, Greece, Spain, Turkey, Ireland.

⁵ The decline in the per capita income coefficient may also be the result of a disequilibrium situation created by the oil price shock. Evidence of this disequilibrium would be the large capital flows between developed, developing and oil producing countries during the post-1973 period.

TABLE 2.—EFFECT OF ENVIRONMENT ON INTER-MEMBER TRADE FLOWS IN REGIONAL INTEGRATION

Integration Scheme	Year	β (1)	$\gamma_1(Y^*/N^*)^2$ (2)	$\gamma_2 D^*$ (3)	Total (1 + 2 + 3) (4)	2 Total (5)
EEC	1970	3.77	3.05	-3.71	3.11	8.63
	1973	4.68	1.74	-3.95	2.47	5.54
	1976	4.83	1.01	-3.30	2.54	5.82
EFTA	1970	3.77	3.06	-4.37	2.46	5.50
	1973	4.68	1.74	-4.56	1.86	3.63
	1976	4.83	1.02	-3.80	2.05	4.14
LAFTA	1970	3.77	2.46	-5.08	1.15	2.22
	1973	4.68	1.46	-5.13	1.01	2.01
	1976	4.83	0.79	-4.27	1.35	2.55
Andean Pact	1970	3.77	2.35	-4.61	1.51	2.85
	1973	4.68	1.41	-4.68	1.41	2.66
	1976	4.83	0.79	-3.95	1.67	3.18
CACM	1970	3.77	2.31	-3.58	2.50	5.66
	1973	4.68	1.30	-3.65	2.33	5.03
	1976	4.83	0.78	-3.04	2.57	5.94

indicates that the effects of integration on trade are diminished as the distance between integrating countries is increased.

The ratio of post- to pre-integration trade is given by $2 \exp(\beta + \gamma_1(Y^*/N^*)^2 + \gamma_2 D^*)$ where Y^* and N^* are the average income and population of the integrating countries and D^* the average distance among them.⁶ This number represents the amount of inter-member trade creation expected in a customs union among countries of a given level of per capita incomes and inter-member distance assuming that the policies adopted to promote integration were of the same effectiveness as that of the average of those adopted by the five integration schemes in our sample. Thus column 5 of table 2 shows the ratio of post- to pre-integration trade expected in each integration scheme had each scheme adopted integration policies of the same effectiveness. The differences shown in column 5 among integration schemes reflect exclusively the impact of environment on the performance of individual integration schemes. Columns 2 and 3 show the separate effects of level of development and of inter-member distance on the effectiveness of integration. Thus, for example, in 1970 the EEC was expected to raise inter-member trade 8.63 fold and the EFTA 5.50 fold. The

expectation of greater inter-member trade in the EEC stems from the fact that the slightly higher per capita incomes in the EFTA cannot compensate for the greater inter-member distances in the EFTA (1,732 miles) compared to the EEC (801 miles).

Overall the differences between the five integration schemes reported in column 2 are relatively small, indicating that differences in per capita incomes explain little of the difference in the ability of integration schemes to stimulate inter-member trade. Rather, the principal environmental source of such differences is, as the data in column 3 suggest, the difference in average inter-member distance, which varies from 306 miles for CACM to 9,173 miles for LAFTA. As a result CACM, although made up of the least-developed countries in our sample, is expected to increase inter-member trade by a factor of five, or somewhat more than is the EFTA. LAFTA, on the other hand, is expected to have a relatively minor impact on inter-member trade largely because of the great distances between members. The decline in γ_1 over time changes the relative impact of environment on integration. With the decrease in the importance of the level of development as a factor promoting inter-member trade in the 1973 and 1976 estimates, developing-country integration schemes are at less of an environmental disadvantage vis à vis developed-country schemes than

⁶ Actually the expression is an upper limit since variance in per capita incomes from the average would reduce the amount of trade creation somewhat.

they are on the basis of the 1970 estimates. Since we have expressed some reservations about the causes of the decline in γ_1 , it would seem more fruitful to view the numbers in column 5 as the range of likely environmental impacts on integration than as point estimates. Nevertheless, the results of table 2 clearly indicate that the effect on inter-member trade that we should expect from a particular integration scheme is strongly influenced by environmental factors and that analyses that ignore such factors are likely to draw incorrect inferences regarding the success of individual integration schemes.

Having thus demonstrated the importance of environmental factors for the effectiveness of regional integration schemes, we turn to an analysis of the policies by which integration has been promoted. Among the five integration schemes in our sample there are important differences in integration policies. Among these differences are the type of integration scheme, such as a free trade area or a common market; the extent to which non-trade barriers are lowered among members; and the height of tariffs imposed on imports from non-members. Our procedure in comparing the effectiveness of integration policies followed by the five integration schemes in our sample is to determine whether the actual increase in inter-member trade is greater than that predicted in table 2. Since increases in trade predicted by table 2 reflect environmental differences between integration schemes but assume identical integration policies for all schemes, any difference between predicted and actual increase in inter-member trade thus reflects differences in the effectiveness of the policies adopted by each integration scheme. The ratio of actual to expected pre-integration trade for the i^{th} integration scheme can be expressed as

$$2^{\beta(i)} = \frac{\text{actual post-integration trade}}{\text{expected pre-integration trade}} \quad (3)$$

$$\beta(i) = (\beta + \pi(i)) + \gamma_1(Y^*/N^*)^2 + \gamma_2 D^* \quad (4)$$

where $\pi(i)$ measures the difference between the effectiveness of the i^{th} integration scheme's policies and the effectiveness of the average integration policy.⁷

⁷ The $\pi(i)$ s are, of course, meaningful only if one accepts the validity of pooling developed and developing country data. Since we were able to carry out only a partial test of the

For three integration schemes, the EEC, LAFTA and the Andean Pact, the $\pi(i)$ s are negative, indicating integration policies of less than average effectiveness. As may be seen from column 4 of table 3, the EEC achieved increases in trade that were equal to 60% of the potential gains that could have been achieved with policies of "average" effectiveness given the characteristics of the integrating countries. LAFTA and the Andean Pact also implemented integration policies of less than average effectiveness. Although the $\pi(i)$ s for these two schemes fluctuate more than those for the EEC, they bracket them, suggesting that integration policies in Latin America were about as effective as those in the EEC and that the differences in trade creation between the EEC and the two Latin American schemes evident in column 1 reflect largely environmental factors. The EFTA and CACM, on the other hand, appear to have implemented policies of above average effectiveness, with those of CACM appearing to be more effective than those of EFTA. Although a discussion relating these findings to the actual integration policies followed in the five schemes is beyond the scope of this paper, we note that the two schemes with more effective policies pursued integration largely for the economic benefits that such integration offered. In contrast, the other three schemes had important non-economic objectives as well. In the EEC these included political unification and the protection of agriculture. In Latin America, the promotion of industry and a reduction in the power of multinational firms were also important objectives of integration.

The problem of the impact of the economic system on the effectiveness of regional integration is somewhat more difficult to deal with than is environment and policy mainly because there is only one socialist integration scheme and all the centrally planned economies in our sample are members. Nevertheless, we can make some rough

validity of such pooling (footnote 2), we also calculated $\pi(i)$ s without resorting to pooling. Equation (2) was estimated for developed and developing countries separately. Then $\pi(i)$ s were calculated for the EEC and EFTA based on parameters obtained from the developed country sample. The $\pi(i)$ s for LAFTA, CACM, and the Andean Pact were calculated using the parameters from the developing country sample. In both cases the relative rankings within the two groups of schemes as well as the magnitude of differences between the $\pi(i)$ s within the two groups remained unchanged from those reported in table 3.

TABLE 3.—EFFECTS OF POLICY ON INTER-MEMBER TRADE FLOWS IN REGIONAL INTEGRATION

Integration Scheme	Year	$\beta(i)$ (1)	$\beta + \gamma_1(Y^*/N^*) + \gamma_2 D^*$ (2)	$\pi(i)$ (3)	$2 \pi(i)$ (4)
EEC	1970	2.35	3.11	-0.76	0.59
	1973	1.73	2.47	-0.74	0.60
	1976	1.77	2.54	-0.77	0.59
EFTA	1970	2.50	2.46	0.04	1.03
	1973	2.34	1.86	0.48	1.39
	1976	1.91	2.05	-0.14	0.91
LAFTA	1970	0.63	1.15	-0.52	0.70
	1973	-0.34	1.01	-1.35	0.39
	1976	0.53	1.35	-0.82	0.57
Andean Pact	1970	1.01	1.51	-0.50	0.71
	1973	-0.27	1.41	-1.68	0.31
	1976	0.91	1.67	-0.76	0.59
CACM	1970	4.00	2.50	1.50	2.83
	1973	3.07	2.33	0.74	1.67
	1976	3.13	2.57	0.56	1.47

estimates of the effect of system on CMEA integration and the efficacy of CMEA integration policies. First, we calculate the environmental impact on intra-CMEA integration policies. This yields a $\beta + \gamma_1(Y^*/N^*)^2 + \gamma_2 D^*$ of 2.47 in 1970, 2.03 in 1973 and 2.22 in 1976. Thus the CMEA has, on the basis of environmental factors, about the same potential for increasing inter-member trade as does the EFTA (see column 4, table 2). The $\beta(i)$ s for CMEA are 1.62, 1.36 and 1.46 for the corresponding years, indicating that CMEA did not, in fact, raise inter-member trade by the amount possible. The $\pi(i)$ s for the three years are -0.85, -0.67, and -0.76, although in this case these numbers capture both the effect of system and policy on inter-member trade. The interesting fact is that the $\pi(i)$ s for CMEA are of roughly the same magnitude as those for the EEC. Thus the combined effect of central planning and of integration policy in CMEA causes that integration grouping to underfulfill its potential by about the same amount as the EEC. Consequently it does not seem that CMEA integration policies or the system of central planning appear to be significantly greater barriers to promoting inter-member trade than do policies adopted among integrating market economies.

IV. Conclusions

We have examined six integration schemes and decomposed their ability to promote inter-member

trade into environmental, policy and system effects. Environmental factors caused the greatest variation in ability to promote trade with inter-member distance being the most important environmental factor. Integration schemes also showed important differences in the effectiveness of their integration policies, with CACM and EFTA having more effective policies than the EEC, LAFTA and the Andean Pact.

On the positive side our findings indicate that effective integration is possible for both developed and developing countries. However, this optimism must be tempered by the realization that for some groups of countries, such as those in Latin America, inter-member distances may severely limit the benefits of integration. Finally, although the combination of policy and system in CMEA has kept that scheme from achieving its full potential for increasing inter-member trade, its performance is quantitatively not different from that of integration schemes among market economies.

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